# Invisible Image Watermarking in the Discrete Wavelet Domain

*A Study with the Daubechies-4 Wavelet*

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**Abstract**

We study invisible digital image watermarking in the wavelet domain. A binary watermark is embedded into a chosen subband of the two-dimensional Daubechies-4 (`db4`) discrete wavelet transform of a grayscale host image by additive perturbation of magnitude $\alpha$, and is extracted non-blindly by re-transforming both the original and the watermarked image. We analyze the scheme along three axes – embedding strength, choice of subband, and robustness under standard attacks (Gaussian noise, Gaussian blur, JPEG compression, resizing, and cropping) – and report quantitative results in mean-squared error, peak signal-to-noise ratio, and normalized correlation. We give a mathematical interpretation of the observed quantization-floor effect at small $\alpha$, and connect our findings to the broader watermarking literature.

**Keywords —** discrete wavelet transform; Daubechies wavelets; digital watermarking; steganography; multiresolution analysis

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# 1 Introduction

## 1.1 Contributions

## 1.2 Roadmap

## 2 Mathematical Background

### 2.1 Multiresolution Analysis

$$\varphi(t) = \sqrt{2}\sum_{k\in\mathbb{Z}} h_k \varphi(2t-k), \quad \psi(t) = \sqrt{2}\sum_{k\in\mathbb{Z}} g_k \varphi(2t-k). \tag{1}$$

### 2.2 The 1-D Discrete Wavelet Transform

### 2.3 The Daubechies Family and `db4`

| Index $k$ | $h_k$ (db4) |
|---|---|
| 0 | $-0.0106$ |
| 1 | 0.0329 |
| 2 | 0.0308 |
| 3 | $-0.187$ |
| 4 | $-0.028$ |
| 5 | 0.6309 |
| 6 | 0.7148 |
| 7 | 0.2304 |

Table 1: Low-pass analysis filter coefficients for the Daubechies-4 wavelet (placeholders – regenerate from `pywt.Wavelet("db4").dec_lo` for exact values).

### 2.4 The 2-D Separable DWT

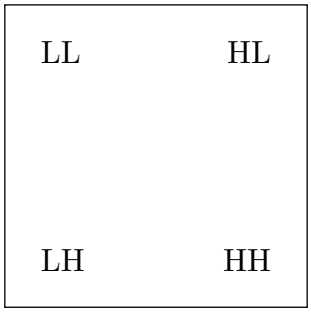



Figure 1: One-level 2-D DWT subband layout (schematic placeholder).

### 2.5 Quality Metrics

$$\mathrm{MSE}\big(I,\hat{I}\big) = \frac{1}{MN}\sum_{i,j}\big(I_{ij}-\hat{I}_{ij}\big)^2, \quad \mathrm{PSNR} = 10\log_{10}\left(\frac{255^2}{\mathrm{MSE}}\right). \tag{2}$$

$$\rho\big(W,\widehat{W}\big) = \frac{\langle W-\bar{W},\ \widehat{W}-\bar{\widehat{W}}\rangle}{\|W-\bar{W}\|\ \left\|\widehat{W}-\bar{\widehat{W}}\right\|}. \tag{3}$$

# 3 The Application: Invisible Watermarking via 2-D DWT

## 3.1 The Watermarking System

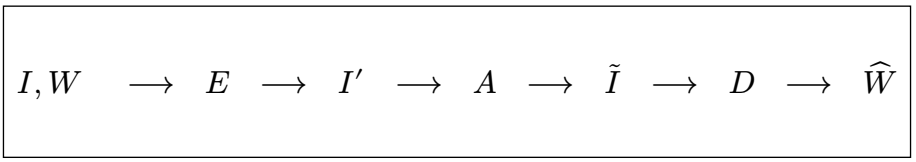



Figure 2: Watermarking pipeline (placeholder for a cetz block diagram).

## 3.2 The Invisibility-Robustness-Payload Trilemma

## 3.3 Why the Wavelet Domain?

## 3.4 Embedding Scheme

$$c'_S = c_S + \alpha\, W, \tag{4}$$

## 3.5 Extraction Scheme (Non-Blind)

$$\widehat{W} = \frac{c'_S - c_S}{\alpha}. \tag{5}$$

## 3.6 Implementation Notes




Figure 3: End-to-end pipeline at the default configuration ($\alpha = 1$, subband `LH`, wavelet `db4`). Left to right: original host image, binary watermark, watermarked image, extracted watermark.

# 4 Experiments and Results

## 4.1 Setup

## 4.2 Experiment 1: Embedding-Strength Sweep

| $\alpha$ | MSE | PSNR (dB) | $\rho$ |
|---|---|---|---|
| 0.01 | 0.494 | 51.19 | 0.634 |
| 0.05 | 0.494 | 51.19 | 0.634 |
| 0.10 | 0.494 | 51.19 | 0.634 |
| 0.50 | 0.494 | 51.19 | 0.634 |
| 1.00 | 0.494 | 51.19 | 0.634 |
| 5.00 | 6.106 | 40.27 | 0.696 |
| 10.0 | 23.627 | 34.40 | 0.696 |
| 20.0 | 93.841 | 28.41 | 0.695 |

Table 2: Embedding-strength sweep on subband `LH`. Source: `results/tables/alpha_sweep_metrics.csv`.

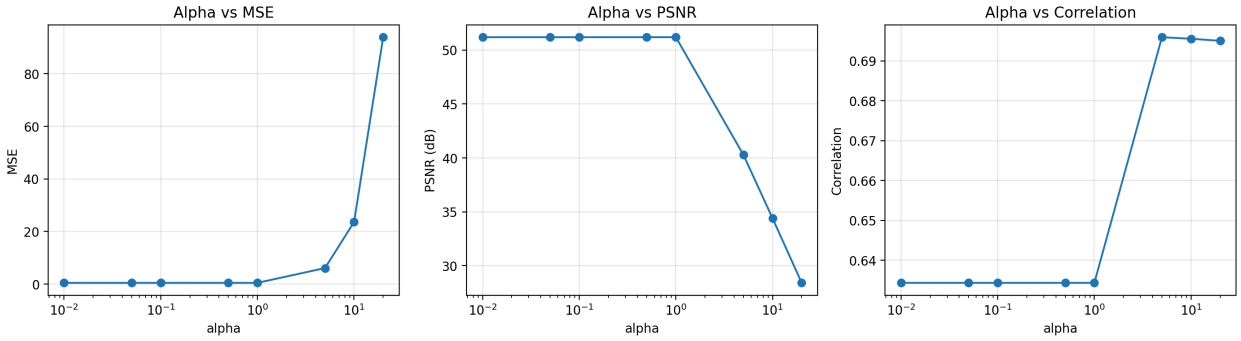
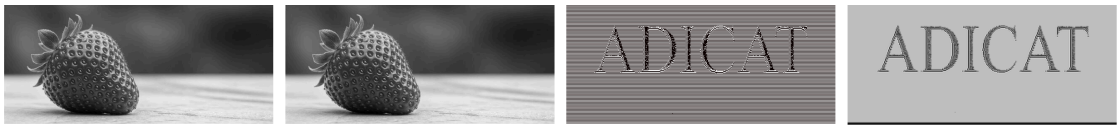



Figure 4: MSE, PSNR, and watermark correlation as a function of embedding strength $\alpha$, on subband `LH` with `db4`. The flat region for $\alpha \leq 1$ is the `uint8` quantization floor analyzed in Section 5.2.

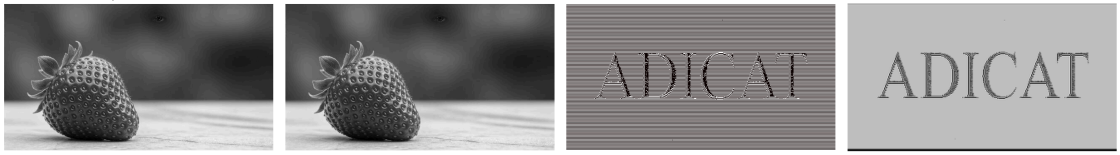



Original
alpha=0.05

Watermarked

Absolute Difference

Extracted Watermark

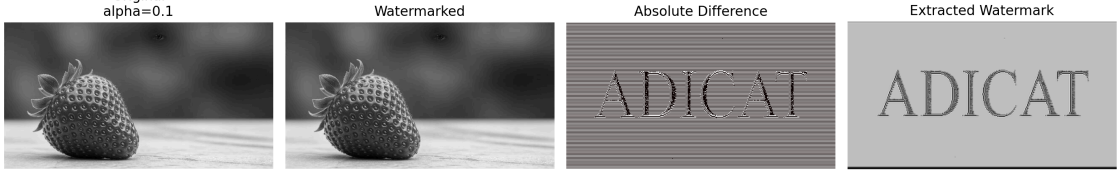



Original
alpha=0.1

Watermarked

Absolute Difference

Extracted Watermark

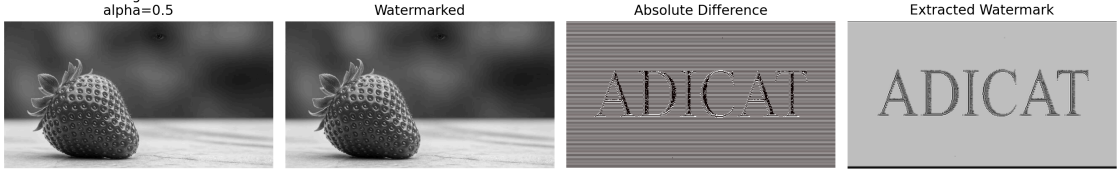



Original
alpha=0.5

Watermarked

Absolute Difference

Extracted Watermark

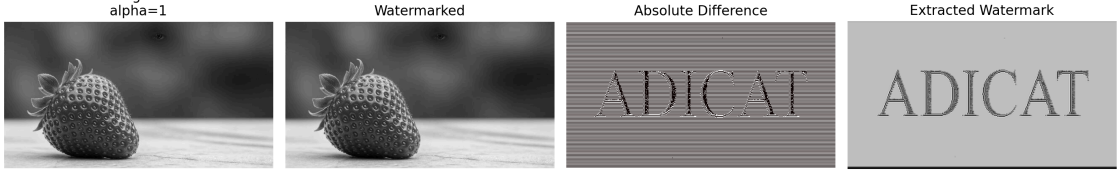



Original
alpha=1

Watermarked

Absolute Difference

Extracted Watermark




Original
alpha=5

Watermarked

Absolute Difference

Extracted Watermark

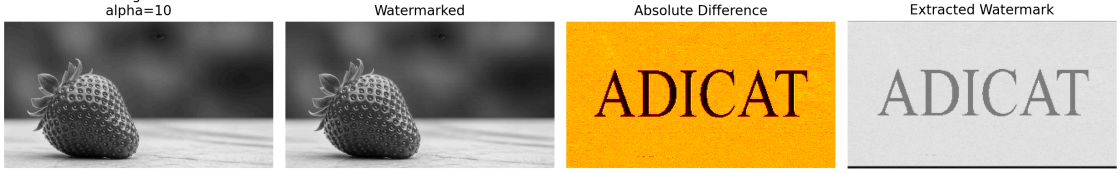



Original
alpha=10

Watermarked

Absolute Difference

Extracted Watermark




Original
alpha=20

Watermarked

Absolute Difference

Extracted Watermark



Figure 5: Visual sweep over embedding strengths. Columns, left to right: original host, watermarked image, absolute pixel difference (hot colormap), extracted watermark. Rows correspond to $\alpha \in \{0.01, 0.05, 0.1, 0.5, 1, 5, 10, 20\}$.

### 4.3 Experiment 2: Subband Comparison

| Subband | MSE | PSNR (dB) | $\rho$ |
|---|---|---|---|
| LL | 3.77 | 42.37 | 0.991 |
| LH | 6.11 | 40.27 | 0.696 |
| HL | 6.11 | 40.27 | 0.796 |
| HH | 6.10 | 40.27 | 0.613 |

Table 3: Subband comparison at $\alpha = 5$. Source: `results/tables/subband_comparison_metrics.csv`.

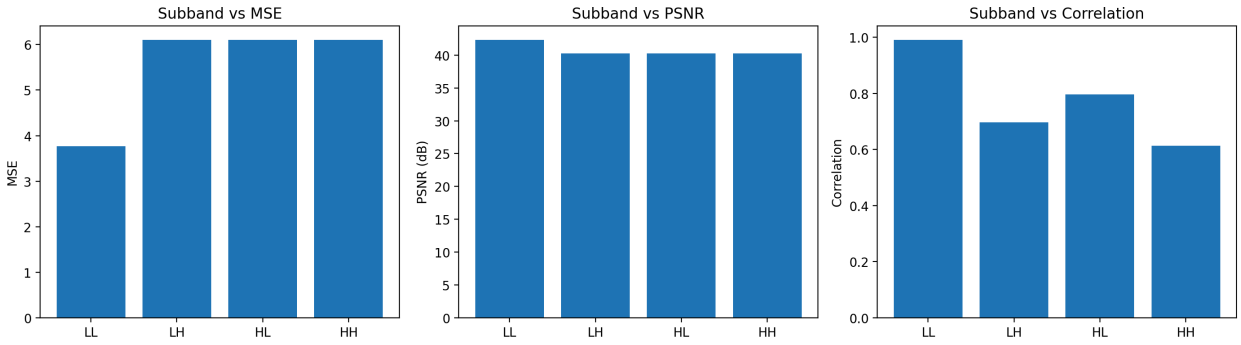



Figure 6: MSE, PSNR, and correlation per subband at $\alpha = 5$ with `db4`.

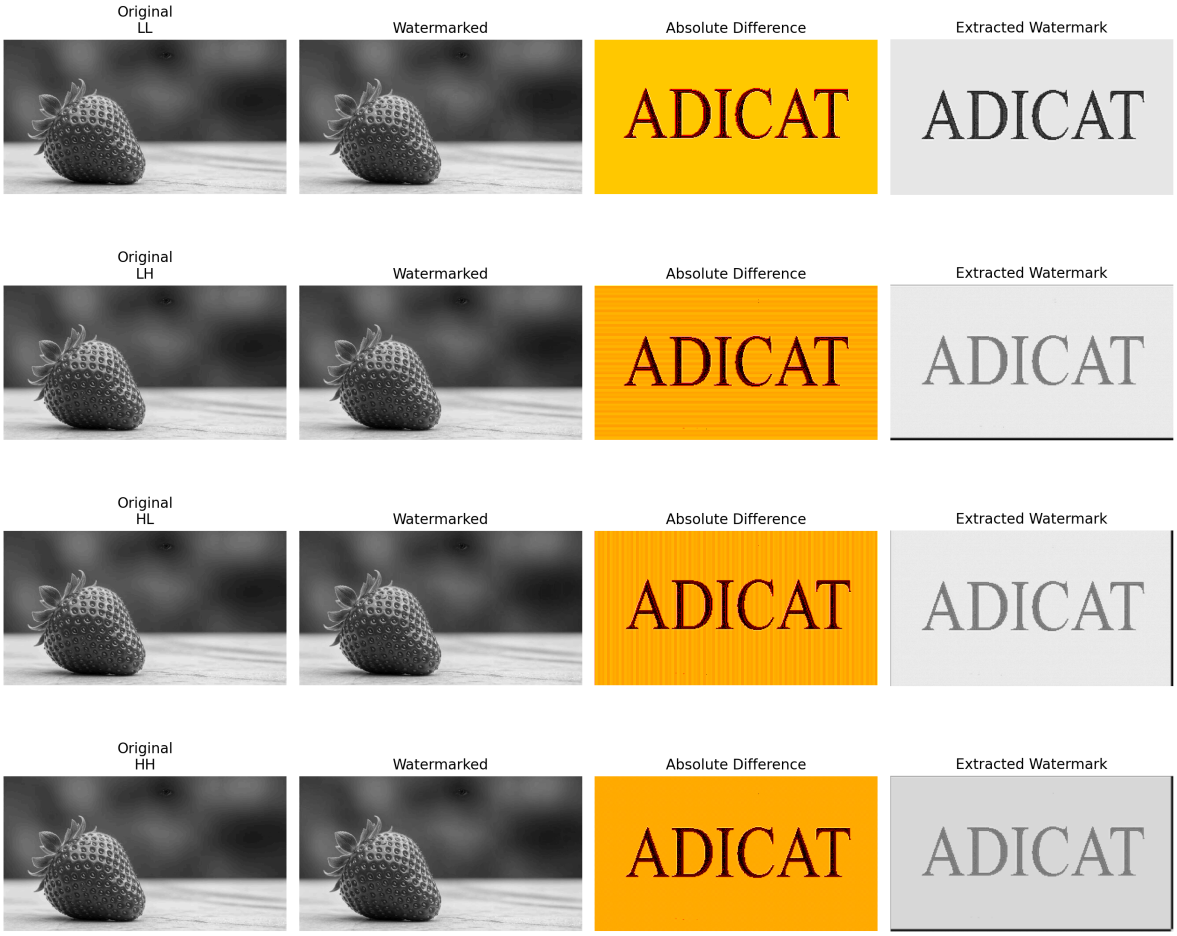



Figure 7: Visual subband comparison. Rows: `LL`, `LH`, `HL`, `HH`. Columns: original, watermarked, absolute difference, extracted watermark.

### 4.4 Experiment 3: Robustness under Attacks

| **Attack** | $\alpha = 1$ | $\alpha = 5$ | $\alpha = 10$ |
|---|---|---|---|
| none | 0.634 | 0.696 | 0.696 |
| Gaussian noise ($\sigma = 8$) | 0.026 | 0.143 | 0.273 |
| Gaussian blur ($5 \times 5$) | −0.006 | −0.005 | −0.004 |
| JPEG ($Q = 40$) | −0.010 | 0.021 | 0.042 |
| resize 60% | −0.002 | 0.001 | 0.004 |
| crop 85% | −0.008 | −0.013 | −0.016 |

Table 4: Watermark recovery correlation $\rho$ after each attack, by embedding strength. Source: `results/tables/attack_robustness_metrics.csv`.

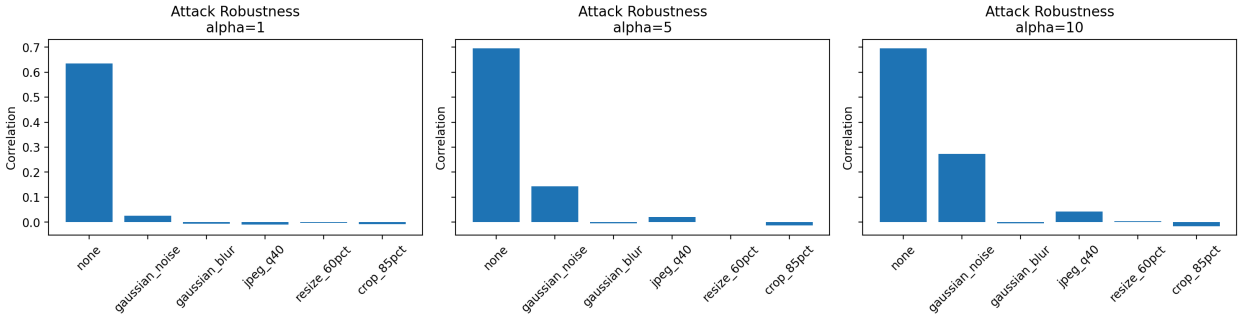



Figure 8: Correlation $\rho$ after each attack, grouped by $\alpha \in \{1, 5, 10\}$.

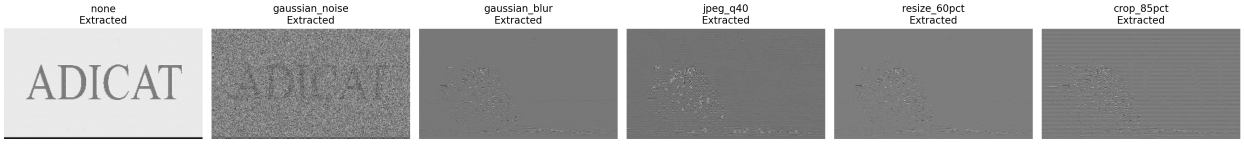



Figure 9: Extracted watermark after each attack at $\alpha = 5$, subband `LH`.

## 5 Discussion

### 5.1 Why `LL` Won the Metric Race

### 5.2 The `uint8` Quantization Floor

### 5.3 Why Blur, JPEG, Resize, and Crop Destroy the Watermark

### 5.4 Limitations

## 6 Conclusions

# A Full Result Tables

## A.1 Alpha sweep

alpha,mse,psnr_db,correlation

0.01,0.49411214953271027,51.192548281480086,0.6343250391468888

0.05,0.49411214953271027,51.192548281480086,0.6343250391468842

0.1,0.49411214953271027,51.192548281480086,0.6343250391468842

0.5,0.49411214953271027,51.192548281480086,0.6343250391468863

1,0.49411214953271027,51.192548281480086,0.6343250391468863

5,6.106003560302626,40.27323307675766,0.6959206607752729

10,23.62684023141967,34.39674716333115,0.6955301514307582

20,93.84058744993324,28.406896430321144,0.6950189043328682

## A.2 Subband comparison

subband,alpha,mse,psnr_db,correlation

LL,5,3.7709390298175345,42.36630850267612,0.9911629157782913

LH,5,6.106003560302626,40.27323307675766,0.6959206607752729

HL,5,6.108455718736092,40.27148930920089,0.7963068671288737

HH,5,6.104890965732087,40.27402449082301,0.6125867694510704

## A.3 Attack robustness

alpha,attack,correlation

1,none,0.6343250391468863

1,gaussian_noise,0.025809556499686352

1,gaussian_blur,-0.006010158406249363

1,jpeg_q40,-0.009523190885764738

1,resize_60pct,-0.0021104319687575354

1,crop_85pct,-0.008037134458404906

5,none,0.6959206607752729

5,gaussian_noise,0.14275217735998397

5,gaussian_blur,-0.005278564937777097

5,jpeg_q40,0.021057974676277593

5,resize_60pct,0.0010100775836045454

5,crop_85pct,-0.01255248775045434

10,none,0.6955301514307582

10,gaussian_noise,0.2729146174760495

10,gaussian_blur,-0.004228783657193783

10,jpeg_q40,0.04200365358945979

10,resize_60pct,0.003575627388871185

10,crop_85pct,-0.01553534356007207

## References